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Deep meta-learning approach for regional parking occupancy prediction considering heterogeneous and real-time information

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ABSTRACT

Parking occupancy prediction enables many smart services in intelligent transportation systems, such as parking guidance and dynamic pricing. Previous studies have improved regional parking occupancy prediction accuracy through many methods, including optimizing model structure and pursuing effective training methods. Moreover, current research realizes that travel behavior is related to multiple sources of external factors, such as spatial relationship and weather. These external factors may not only be real-time information that can be expressed numerically, but also a state that can have a long-term impact and is difficult to represent with time-series information. It remains a challenge to consider both these temporal and heterogeneous information in the network. To fill the research gap, we propose a spatio-temporal parking occupancy prediction approach incorporating heterogeneous and real-time weather as external auxiliary data, named Meta-HWFST, which leverages the integration of innovative global-adjacent dual-attention, multivariate decoder and model-agnostic meta-learning pretraining. The proposed approach is evaluated on a real world dataset of 58 parking lots. The results empirically showed that Meta-HWFST achieved an average improvement of 32.45% in accuracy, compared to other representative models. We conduct a comprehensive ablation experiment to demonstrate the contribution of each module to the improvement. Moreover, we demonstrate that our model could reasonably assign weights through feature importance analysis.

1. Introduction

With the proliferation of automobiles, the supply of parking spaces can hardly meet the demand. Studies have shown that it takes an average of 3.5–14 min per trip to find a parking space because of the lack of parking spaces, resulting in a 25%–40% increase in peak traffic and more serious pollution [1,2]. An unlimited supply of automobile parking spaces is impossible, and more and more managers are turning to smart parking management such as dynamic pricing, space sharing, and parking recommendations [3,4]. As a foundation to reveal future parking demand and enable these intelligent services, regional parking occupancy prediction have attracted increasing research attention [5,6].

In early studies, statistics-based regression methods were widely used for parking behavior prediction. These methods are simple and rapid to calculate. However, they are limited to model intricate non-linear relationships, especially spatio-temporal ones [7–9]. With the

improvement of computer science, machine learning and neural networks with powerful feature extraction capability have become research hotspots [10]. Previous deep learning spatio-temporal prediction studies have typically used graph neural networks (e.g., Graph convolutional networks (GCN)) to model adjacency relationships and recurrent neural networks (e.g., Long short-term memory (LSTM) and Gated recurrent unit (GRU)) to model temporal relationships, respectively [11–13]. The introduction of attention mechanism (including graph attention and temporal attention) allows the weights of the network to be explicitly determined and the prediction accuracy is further improved [14,15]. However, previous graph-based approaches primarily focus on neighboring information [16]. They ignore the fact that functionally similar parking lots can also provide useful information to each other, even if they are not adjacent. For example, parking lots located in different shopping centers have more similar patterns of occupancy changes during holidays than their hospital-specific neighbors [17,18].

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Moreover, unlike traffic flow nodes, not every parking lot in a regional parking dataset has other nodes in a given surrounding. In traditional traffic flow prediction tasks, researchers would achieve long-range node information propagation by deepening the graph network or by allowing the information to wander randomly across the graph. In such a case, no matter how many layers of graph structure we embed in the network, discrete nodes will always be isolated in information propagation process. Therefore, neighborhood-based graph modeling is not sufficient for accurate regional parking occupancy prediction and modeling global data dependencies is a challenge.

Intelligent information systems enable government departments to efficiently integrate multiple sources of information, which makes it possible to transform urban management from a single perspective to those with multiple influencing factors. Parking behavior is affected by complex environmental factors, and taking these factors into account can reduce lags and improve accuracy [19,20]. Among them, weather has become one of the most popular auxiliary information for parking prediction tasks due to its significant impact on people's travel behavior. Most previous studies have simply concatenated real-time sensor data related to parking behavior (e.g., temperature, rainfall and humidity) with occupancy data and then used convolutional networks for data fusion [21]. The convolutional approach has an improved but mediocre effect on prediction accuracy. A study shows that using a convolutional approach to consider the effect of weather on parking behavior leads to just a 1.1%–1.8% improvement [22]. It is not sufficient to provide the model with weather information equal to the length of the occupancy sequence. For example, travelers are more inclined to stay home if they know that it is likely to rain on that day. It is also a challenge to fuse heterogeneous weather information over long time periods (e.g., a whole day) to inform accurate predictions.

In summary, there are still challenges in regional parking occupancy prediction as listed below.

- Global correlation perception: Traditional neighborhood-based graph learning is not sufficient for accurate regional parking occupancy prediction. There is still a lack of research on how to adaptively and quantitatively evaluate and model the potential relevance of parking lots, especially for non-adjacent parking lots.
- Heterogeneous factor fusion: The perceived length of influences (e.g., weather) has been inadequate in previous studies. In smart city management where multiple heterogeneous information can be aggregated, predictions that can efficiently consider the long-term effects of heterogeneous information are still lacking.
- Network structure optimization: To achieve accurate and efficient parking occupancy prediction, we still need to explore suitable real-time information learning models.

To address the above challenges and achieve accurate prediction, a data driven approach, named Meta-HWFST, is proposed. The following three modules are contained. (1) Global-adjacent dual-attention module. Global attention in different weather situation are computed to determine how helpful parking lots globally are for local occupancy predictions, thereby enhancing model global fusion capabilities. Furthermore, graph attention mechanism is employed to learn information from neighboring parking lots. (2) Multivariate decoder module. This module enables LSTMs in different lengths, temporal attention mechanisms and multilayer perceptrons to be incorporated as multivariate decoder of the stacked high-dimensional features and temporal data (e.g. temperature). (3) Model pre-training module. To further improve prediction accuracy, a Model-Agnostic Meta-Learning pre-training framework, Reptile [23], is introduced. Different weather is used as a basis for grouping the data. The grouping training and gradient descent of meta-learning enables the model to better learn the differences between the weather without losing the ability to learn the commonalities.

Performance of the proposed approach Meta-HWFST is evaluated through a dataset containing 58 parking lots, which is sampled from

Guangzhou, China, from June 1 to June 30, 2018. The results empirically show that Meta-HWFST obtains the state-of-the-art performance in all the four evaluation metrics. Ablation experiment shows that the proposed global attention and meta-learning pre-training bring a substantial enhancement.

The main contributions of this paper are as follows.

- We propose a data driven approach that addresses the challenge of spatio-temporal parking occupancy prediction incorporating heterogeneous and real-time information. The model contains a global-adjacent dual-attention module, an improved multivariate decoder module, and a model pre-training module.
- We propose a scheme for effectively consider the situation of weather over a long period, which is a heterogeneous information. We propose a method to compute global attention under different weather to provide different auxiliary information for prediction. Moreover, model-agnostic meta-learning is developed to learn patterns under different weather. It can help to learn the identity of different categorical data without losing commonality.
- We propose a multivariate decoder module for high-dimensional features and isomorphic data fusion. LSTMs with different lengths are used to learn hierarchical temporal patterns, and temporal attention is used to strengthen the network and avoid ambiguity.

The rest of the paper is summarized below. Section 2 is a literature review that presents some researches on parking occupancy and traffic prediction and summarizes the strengths and weaknesses of the researches. Section 3 is the methodology, which provides an introduction to the proposed approach. The approach presented in Section 3 is validated in detail in Section 4 with a dataset of 58 parking lots. Finally, Section 5 concludes the work and discusses future research directions.

2. Literature review

Accurate urban parking occupancy prediction provides the groundwork for intelligent transportation systems and information dissemination systems [20]. Early time series prediction models include historical average model (HA) [24], Auto Regressive Integrated Moving Average Model (ARIMA) [7], and Least absolute shrinkage and selection operator (Lasso) [8]. These models calculate the future status based on selected variables and corresponding coefficients. A real-time availability forecasting algorithm is proposed that assigns simulated parking requests, estimates future departures and ultimately predicts parking occupancy [25]. A prediction model based on a time-space dependent multivariate autoregressive model is proposed to predict the short-term parking occupancy in a single parking lot [26]. A continuous-time Markov M/M/C/C queue model is employed to predict parking occupancy for various prediction intervals based on historical occupancy data [9]. However, statistical regression methods have difficulty modeling complex features in time series, resulting in poor prediction accuracy. Moreover, statistical models are difficult to model spatial relationships and are not conducive to developing joint predictions in an area.

In recent studies, the powerful feature extraction capabilities of machine learning have been utilized to improve the prediction accuracy of urban parking occupancy [27]. Nowadays, neural networks with non-linear pattern learning ability are rapidly becoming a research hotspot. Recurrent neural network and its variants are already a well-established network component in processing time-series information [12,13,28]. A novel multistep long short-term memory recurrent neural network (LSTM-NN) model is proposed to predict parking occupancy based on historical parking occupancy information [1]. Baidu proposed a parking occupancy prediction model, Du-parking, incorporating LSTM and linear layer output, which is successfully applied on Baidu Maps App to provide commercial services in major cities such as Beijing and Shenzhen [10]. While RNNs can capture the temporal correlation of sequences, they ignore the spatial correlation of nodes.

Spatio-temporal prediction studies mainly used convolutional methods to consider neighboring features in early studies. In a parking occupancy prediction model, temporal and spatial correlations are captured by the two parallel ConvLSTM, and meanwhile, the dense convolutional network (DCN) is leveraged to further improve the propagation and reuse of features in the prediction process [29]. A traffic prediction model, STGCN, which builds spatio-temporal models with convolutional structures, achieves state-of-the-art on several tasks [30].

Convolutional networks are inherently suitable for Euclidean spaces but not for complex traffic topologies. Classical convolutional methods cannot directly handle non-Euclidean data, and graph-based methods solve this problem [31]. Graph Convolutional Neural Networks (GCN) are first proposed to consider graph structural features in a convolutional way [11]. In a spatio-temporal parking occupancy prediction model, GCN and GRU are integrated to consider temporal parking records, similarities, and spatial correlations of parking lots [32]. A graph-based model ST-GBGRU is proposed, which is capable of simultaneously extracting spatio-temporal features [33].

Attention mechanisms have many benefits for neural network, avoiding convolutional ambiguity and recurrent obliviousness. As important components of time series research, temporal attention mechanisms and Transformer have long been recognized [15,34]. An attention module named distAtt is proposed together with a spatio-temporal convolutional block to mix the long-term spatio-temporal correlations and instantaneous ones [35]. A hierarchical graph convolution module is proposed to model spatial correlations among parking lots with attention mechanism [36]. However, current research still focuses on the extraction of adjacency information, ignoring the similar change patterns of non-neighboring nodes. There is still a lack of research on how to take global data correlation into account, especially using attention mechanisms to determine the magnitude of global attention.

Moreover, there are many parking-related factors that affect the prediction accuracy, such as weather, temperature, rainfall and traffic flow [4,37,38]. For example, if inclement weather conditions occur on weekdays, travelers will be delayed due to slow traffic, which also affects parking occupancy over time. Therefore, to predict parking occupancy more effectively, it is necessary to model the correlation between multiple sources of data. A model capable of taking multiple heterogeneously structured traffic data as input is proposed, including parking meter transactions, traffic speed, and weather conditions [22]. A novel model is proposed, which predicts parking behavior based on historical records, weather, environments, weekdays, and events [39]. In a data fusion model, the attributes set of the category factors (e.g., day of week and weather conditions) and the time information (e.g., time of day) are included and integrated into the parking occupancy prediction process [40]. MePark system extracts implicit features from multiple sources of data, including time of day, road network, POIs, weather, and holiday events [41]. In this system, features with different characteristics and data formats are modeled by three different types of components. In terms of commercial applications, a real parking system, ThsParking, developed by Huaching Technology, combines influencing factors such as weather and periodicity to achieve accurate predictions. Based on the prediction results, the parking system can adopt a segmented pricing strategy to better utilize the parking spaces and make more profit [42]. It can be learned that modeling the data related to parking behavior is necessary for parking occupancy prediction tasks. However, most previous studies have simply concatenated multi-source data with parking data, which improves prediction accuracy but not significantly. Moreover, how to incorporate heterogeneous external data (e.g., weather over an extended period of time) remains a challenge.

Allowing artificial intelligence systems to “learn to learn” is an important way to achieve high efficiency. Meta-learning allows neural networks to fully incorporate knowledge from past tasks and learn quickly about new tasks [43]. Therefore, it provides new ideas for considering different environmental status (e.g., weather). Meta-learning

approaches include Model-agnostic Meta-learning (MAML) [44] and its extension, First-Order Model-agnostic Meta-learning (FOMAML) [45] and Reptile [23]. The model parameters are trained explicitly so that only a small number of gradient steps and training data for a new task are required to obtain good generalization performance. In a single parking lot occupancy prediction task, FOMAML is applied to fasten the adaptation and convergence of the model through a pre-training process [18]. Different weather conditions can be regarded as different training subtasks, and meta-learning can be exactly an efficient way to deal with them, but such an approach still lacks research.

In summary, neural networks that can incorporate spatio-temporal features and multiple influencing factors provide methodology for accurate parking occupancy prediction. Moreover, the introduction of attention mechanisms allows spatio-temporal relationships to be better modeled. But non-adjacent data relationships remains to be studied. Parking-related data (e.g., weather and temperature) has proved to be helpful in improving the accuracy of prediction. However, it remains a challenge to efficiently fuse various isomorphic and heterogeneous external auxiliary information. Meta-learning that allows pre-training of data groups provides a new idea for learning about patterns in different situation (e.g., different weather scenarios). Therefore, to tackle the challenges and achieve accurate parking occupancy prediction, we propose a heterogeneous weather fusion meta-learning approach with effective global-adjacent dual attention.

3. Methodology

As illustrated in Fig. 1, the proposed model consists of three modules, namely (a) a global-adjacent dual-attention module, which incorporates global attention under different weather scenarios and adjacency graph attention to learn global data correlations and neighborhood features, respectively; (b) a multivariate decoder module, which develops LSTMs of different lengths and temporal attention to extract hierarchical temporal features and multilayer perceptrons to fuse multiple high-dimensional features and temporal auxiliary information; (c) a model pre-training module, which introduce a “learning to learn” method, Reptile, to efficiently learn the features between data classifications in different weather.

3.1. Problem definition

Parking occupancy is defined as the number of cars being serviced in a specific parking lot within a given time interval (e.g., 5 min). Parking occupancy in all parking lots in an area is defined as $\{O_1, O_2, \dots, O_t, \dots\} \in \mathbb{R}^{N_p \times N_T}$, where N_p is the number of parking lots, and N_T is the number of time steps. Demand at the t -th moment is defined as $O_t = \{O_{1,t}, O_{2,t}, \dots, O_{N_p,t}\}$, where $O_{N_p,t}$ denotes the parking occupancy in the N -th observation parking lot at the t -th moment. To support the training of data-driven model, the sliced demand data with a window size τ is used, which is denoted as $\{O_{t-\tau+1}, O_{t-\tau+2}, \dots, O_t\}$. Similarly, auxiliary temporal data (e.g., temperature) of all parking lots is defined as $P = \{U_1, U_2, \dots, U_t, \dots\} \in \mathbb{R}^{N_p \times N_T}$, and sliced auxiliary data is denoted as $\{U_{t-\tau+1}, U_{t-\tau+2}, \dots, U_t\}$. To further improve prediction accuracy, the proposed model requires to learn the adjacency features, which is represented by an undirected graph G . We also categorize data over time by weather conditions; for example, in southern China, where the magnitude of rainfall can have a significant impact on travel behavior, we classify the data into three weather groups: no rain, light rain, and heavy rain. The regional parking occupancy prediction problem can be formulated by Eq. (1).

$$\{O_{t-\tau+1}, O_{t-\tau+2}, \dots, O_t, U_{t-\tau+1}, U_{t-\tau+2}, \dots, U_t, G\} \xrightarrow{f} \hat{O}_{t+\tau} \quad (1)$$

where $\hat{O}_t$ denotes the predicted demand at the t -th moment.

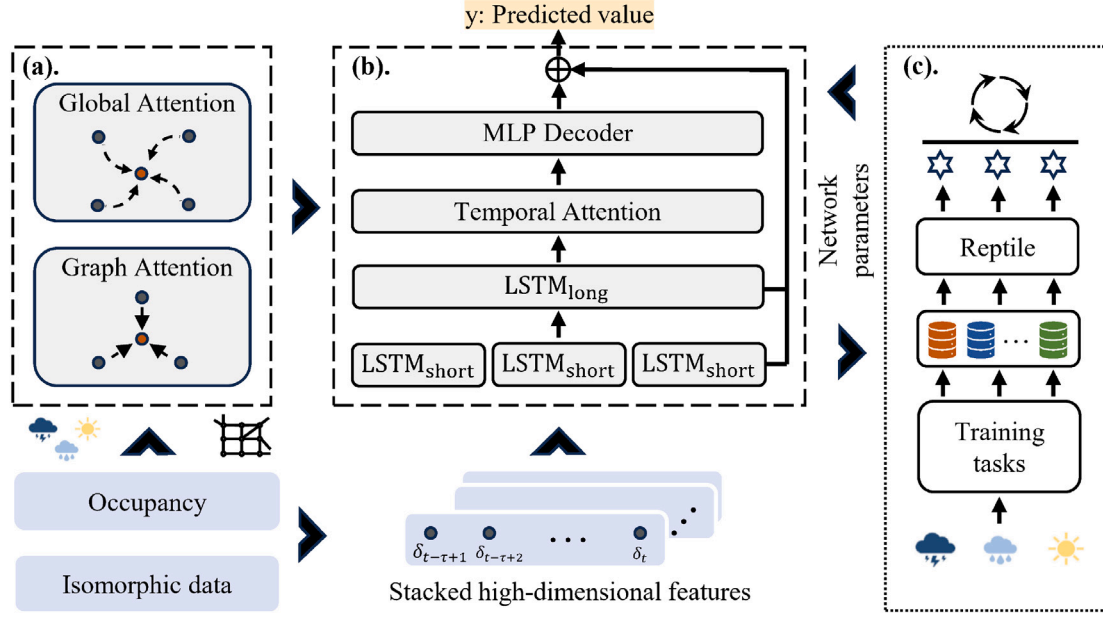


Fig. 1. The model structure of the proposed approach, which consists of (a). Global-adjacent dual-attention module, (b). Multivariate decoder module, and (c). Model pre-training module.

3.2. Global-adjacent dual-attention module

3.2.1. Global attention

Parking behavior has a strong relationship with other services around the parking lot. For example, the parking lots of two commercial facilities have similar parking peaks during their daytime business hours, even though the two parking lots are not geographically close to each other. Previous research in spatio-temporal traffic flow prediction usually involves multiple layers of graph neural networks to achieve information dissemination over long distances. However, such an approach is not applicable to outlying parking nodes in urban areas. To focus on global feature similarity, an attention method based on the similarity of the training set is proposed. In particular, since weather conditions have different impact on travel behavior, the calculation of feature similarity is processed based on the classification of weather conditions of the day (e.g., no rain, light rain, and heavy rain). We determined the correlation between the parking lots through the Pearson correlation coefficient, as shown in Eq. (2). Then, I is introduced so that positively correlated parking lots are assigned attention in the computation of softmax, as shown in Eq. (3). The attention matrix A_f is calculated based on the global feature similarity as Eq. (4).

$$\text{Corr}_{ij} = \frac{\sum_{t=1}^n (x_t^i - \bar{x}^i) (x_t^j - \bar{x}^j)}{\sqrt{\sum_{t=1}^n (x_t^i - \bar{x}^i)^2} \sqrt{\sum_{t=1}^n (x_t^j - \bar{x}^j)^2}} \quad (2)$$

$$I_{ij} = \begin{cases} \text{Corr}_{ij}, & \text{if } \text{Corr}_{ij} \geq 0 \text{ and } i \neq j \\ -\infty, & \text{else} \end{cases} \quad (3)$$

$$A_{fij} = \text{softmax}_j(I_{ij}) = \frac{\exp(I_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(I_{ik})} \quad (4)$$

Then, the operation as in Eq. (5) is performed, where MLP is a multilayer perceptron, O is the input occupancy data, and ξ is the output hidden state of global attention. Note that this module does not process information in temporal dimension. We introduce a structure similar to residual connections in the process of global attention computation, where α is a hyperparameter.

$$\xi = \text{MLP}((1 - \alpha)A_f O + \alpha O) \quad (5)$$

The MLP contains three layers of linear cells, where the activation function is Rectified Linear Unit (ReLU) [46]. ReLU is shown in Eq. (6). The model's MLP decoder is the same.

$$\text{ReLU} = \max(0, x) = \begin{cases} x, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases} \quad (6)$$

3.2.2. Adjacent attention

Well-developed road networks and intelligent information dissemination systems allow parking demand to be diverted to neighboring parking lots when a specific parking lot is in high demand. We consider the neighborhood relations as a graph structure. The modeling process of the graph is shown in Eq. (7), where ϖ_{ij} stands for the connection state of the i th and j th parking lot, and Dist_{ij} stands for the driving distance of the two parking lots. Parking lots within driving distance δ are considered to be spatially connected.

$$\varpi_{ij} = \begin{cases} 1, & \text{if } \text{Dist}_{ij} \leq \delta \\ 0, & \text{if } \text{Dist}_{ij} > \delta \end{cases} \quad (7)$$

Graph attention mechanism (GAT) is used to model adjacencies. The input of GAT is the occupancy matrix O . Note that like global attention, adjacent attention does not process information in temporal dimension. The computing processes are divided into two steps. The first step is attention score ω computation as Eq. (8).

$$\omega_{ij} = \text{LeakyReLU}(W(O_i \parallel O_j) + b) \quad (8)$$

where W is the parameter matrix of the fully connected network, b is the bias, O_i and O_j denote occupancy of the i th and j th parking lot, LeakyReLU [47] is a nonlinear activation function as Eq. (9), and " $\parallel$ " means vector concat.

$$\text{LeakyReLU} = \begin{cases} x, & \text{if } x \geq 0 \\ 0.01x, & \text{if } x < 0 \end{cases} \quad (9)$$

The second step is hidden status update. For each node on the graph, a softmax function is used to calculate the attention values A_{Gij} of all its neighbors, as Eq. (10).

$$A_{Gij} = \text{softmax}_j(\omega_{ij}) = \frac{\exp(\omega_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(\omega_{ik})} \quad (10)$$

where $\mathcal{N}_i$ stands for the set of i th node's neighbors.

The output hidden status of i th node is shown in Eq. (11), including the influence of surrounding nodes on this node.

$$\mathbf{r}_i = \text{ReLU}(\sum_{j \in \mathcal{N}_i} \mathbf{A}_{G_{ij}} \mathbf{O}_j) \quad (11)$$

To strengthen the spatial information representation, we introduce multi-head attention to graph attention. Multi-head attention mechanism means that the attention model is divided into multiple heads, forming multiple subspaces that allow the model to attend to different aspects of information. The calculation here is as shown in Eq. (12).

$$\mathbf{r}_i = \sum_{k=1}^K \text{ReLU}(\sum_{j \in \mathcal{N}_i} \mathbf{A}_{G_{ij}}^k \mathbf{O}_j) \quad (12)$$

The output of adjacent attention is $\mathbf{R} = [\mathbf{r}_1, \dots, \mathbf{r}_{N_p}]$, where $\mathbf{r}_i$ stands for the output hidden state of the i th node.

3.3. Multivariate decoder module

The high-dimensional hidden states ξ and $\mathbf{R}$ output by the global-adjacent dual-attention module are used as input to this module. Moreover, we provide the interface for other external isomorphic information fusion. For example, temperature data from weather stations are considered as time-series data related to travel behavior. External isomorphic information and high-dimensional self-information are first stacked together and then used as feature inputs to the network.

Previous models use a single temporal network to perceive the full length of temporal information, which may bring about a lack of attention to short-term features. An improved temporal network, which can learn hierarchical patterns in time series, is proposed. It contains $\text{LSTM}_{\text{short}}$ for learning short-term features and $\text{LSTM}_{\text{long}}$ for learning long-term features. The last outputs of both LSTMs are used as two of the outputs of the module. Besides, the self-attention mechanism is introduced to assign weights rationally in the temporal dimension.

LSTM is an improved recurrent neural network that achieves excellent performance on temporal prediction tasks. It defines a recurrent function F_L and calculates $\mathbf{h}_t$ for time t , as Eq. (13).

$$\mathbf{h}_t, \mathbf{s}_t = F_L(\mathbf{h}_{t-1}, \mathbf{s}_{t-1}, \mathbf{d}_t) \quad (13)$$

where $\mathbf{h}_t$ is the hidden layer state of the LSTM at time t , $\mathbf{s}_t$ is the final state of memory cell at time t , and $\mathbf{d}_t$ is the value of feature sequence at time t .

The calculation is defined by Eq. (14):

$$\begin{aligned} \mathbf{i}_t &= \text{sigmoid}(\mathbf{W}_{d_i} \mathbf{d}_t + \mathbf{W}_{h_i} \mathbf{h}_{t-1}) \\ \mathbf{f}_t &= \text{sigmoid}(\mathbf{W}_{d_f} \mathbf{d}_t + \mathbf{W}_{h_f} \mathbf{h}_{t-1}) \\ \mathbf{o}_t &= \text{sigmoid}(\mathbf{W}_{d_o} \mathbf{d}_t + \mathbf{W}_{h_o} \mathbf{h}_{t-1}) \\ \mathbf{s}_t &= \mathbf{f}_t \odot \mathbf{s}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_{d_g} \mathbf{d}_t + \mathbf{W}_{h_g} \mathbf{h}_{t-1}) \\ \mathbf{h}_t &= \mathbf{o}_t \odot \tanh(\mathbf{s}_t) \end{aligned} \quad (14)$$

where t is the time step, $\mathbf{i}_t$ is input gate, $\mathbf{f}_t$ is forget gate, $\mathbf{o}_t$ is output gate, $\mathbf{i}_t$, $\mathbf{f}_t$, and $\mathbf{o}_t \in \mathbb{R}^m$. $\mathbf{W}_{d_i}$, $\mathbf{W}_{d_f}$, $\mathbf{W}_{d_o}$, $\mathbf{W}_{d_g}$, $\mathbf{W}_{h_i}$, $\mathbf{W}_{h_f}$, $\mathbf{W}_{h_o}$ and $\mathbf{W}_{h_g}$ are coefficient matrix. $\mathbf{W}_{d_i}$, $\mathbf{W}_{d_f}$, $\mathbf{W}_{d_o}$ and $\mathbf{W}_{d_g} \in \mathbb{R}^{m \times n}$, m, n are the dimension of variable. $\mathbf{W}_{h_i}$, $\mathbf{W}_{h_f}$, $\mathbf{W}_{h_o}$ and $\mathbf{W}_{h_g} \in \mathbb{R}^{m \times m}$, and $\odot$ denotes elements-wise multiplication.

The structure of the proposed module is shown in Fig. 2. First, the hidden matrix is divided into three parts. Each part is processed by an $\text{LSTM}_{\text{short}}$. The last output of the last $\text{LSTM}_{\text{short}}$ in time order will be used as one of the outputs of the module. Then, the output hidden states of $\text{LSTM}_{\text{short}}$ are concatenated as the input of $\text{LSTM}_{\text{long}}$. Similarly, the last output of the $\text{LSTM}_{\text{long}}$ will be used as one of the outputs of the module.

The output of $\text{LSTM}_{\text{long}}$ is processed by three linear modules. $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ are obtained respectively. As shown in Eq. (15), $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ is used for Scaled Dot-Product Attention to obtain the attention score on the temporal sequence.

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}) \mathbf{V} \quad (15)$$

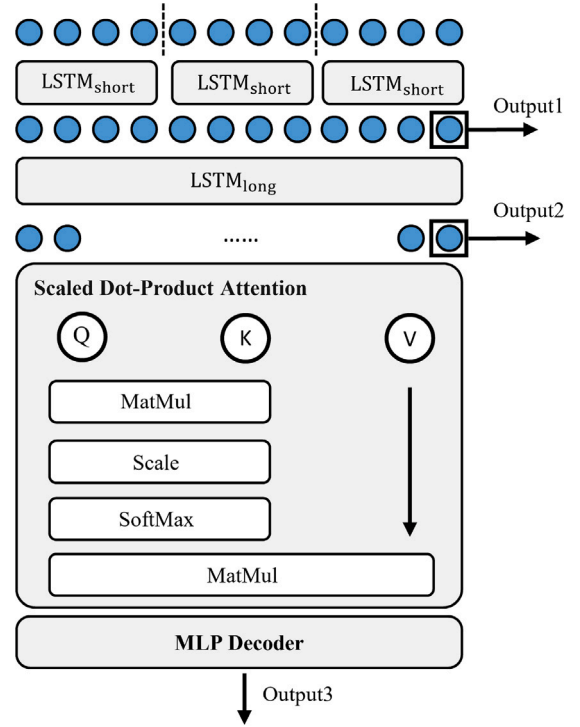


Fig. 2. The structure of the proposed temporal module.

where $\sqrt{d_k}$ is a scaling factor.

A multilayer perceptron (MLP) is used as a multivariable decoder. Same as global attention, The MLP contains three layers of linear cells, where the activation function is Rectified Linear Unit (ReLU). Finally, the three outputs are added together as the predicted value.

3.4. Model pre-training module

Previous approaches typically allowed only isomorphic dynamic information inputs, making it difficult to account for heterogeneous weather in the model. Thus long periods of implicit weather effects are lacked consideration. To efficiently learn different patterns of parking behavior under different weather conditions separately, an improved version of MAML, Reptile, is introduced to obtain an optimal initialization parameter for the model. The meta-training set will be divided into some parts according to the weather report of the day. The grouping training and gradient descent of meta-learning allows the model to better learn the differences between weather without losing the ability to learn commonalities. In contrast, ordinary training methods will average the differentiated features when calculating the total loss.

Given that M defines the pre-training tasks, the objective function of MAML and Reptile, which is to find a set of initial parameters minimizing the learner loss, can be defined by Eq. (16), where θ is the parameter related to the initial parameter ϕ ; θ_m represents the parameter in task m ; $l^m(\theta^m)$ is the error with θ^m in task m , which is calculated by the loss function of the model; and $L(\phi)$ is the total after-training loss of initial parameter ϕ .

$$\min L(\phi) = \sum_{m=1}^M l^m(\theta^m) \quad (16)$$

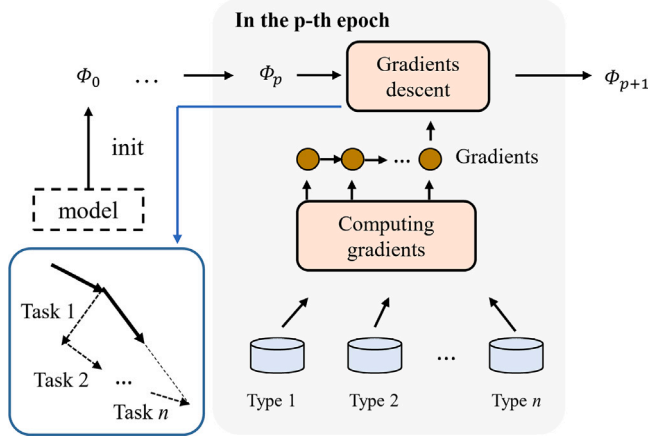
Reptile is an algorithm closely related to First-Order Model-Agnostic Meta-Learning (FOMAML), which is more simple to implement than MAML. Unlike MAML and FOMAML, Reptile does not need a support-query split for each task. The calculation process of Reptile is shown in Algorithm 1.

Algorithm 1 Reptile Algorithm for Meta-HWFST**Require:** $p(\mathcal{T})$: distribution over tasks**Require:** η_o, η_i : the outer and the inner step size hyperparameters

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1: randomly initialize  $\theta$ 
2: while not done do
3:   Sample batch of tasks  $\mathcal{T}_i \sim p(\mathcal{T})$  of different types of weather
4:   for  $\mathcal{T}_i$  do
5:     Evaluate  $\nabla_{\theta} l^i(\theta^i)$ 
6:     Compute adapted parameters with gradient descent:  $\theta'_i = \theta - \eta_i \nabla_{\theta} l^i(\theta^i)$ 
7:   end for
8:   Update  $\theta = \theta + \eta_o \sum_i^M (\theta_i - \theta)$ 
9: end while

```

**Fig. 3.** The iterative processes of Reptile in our tasks.

In Reptile, only a first-order gradient update is required, which saves a lot of computational cost. At the same time, Reptile is theoretically capable of achieving the effectiveness and accuracy of MAML. The iterative processes and the gradient descent processes of Reptile in our tasks are shown in Fig. 3.

4. Experiments

In this section, the proposed model is compared with the current state-of-the-art prediction model. To ensure fairness, the experiments will use the same evaluation metrics and try to maintain consistent hyper-parameter settings. In addition, ablation experiments will be conducted to demonstrate the contribution of each part of the model in improving the prediction accuracy. Besides, a noise impulse response experiment is introduced to illustrate the importance of each feature.

4.1. Data description

4.1.1. Occupancy data of parking lots

The proposed method is validated on a real dataset, which is collected from Tianhe District, Guangzhou, China, with a sampling interval of 5 min, from June 1 to 30, 2018. As shown in Fig. 4, the dataset covers 58 parking lots in the whole Tianhe District of 137.38 km², including 29 parking lots of commercial facilities, 21 parking lots of office buildings, and 8 parking lots of residential areas. It can be seen that some parking lots that are discrete, i.e. they are not adjacent to other parking lots, which is a difficult problem in spatio-temporal prediction. Discrete nodes do not pose a problem for traditional spatio-temporal model inputs because the data structure of the data matrices and graph matrices are no different from those of traditional spatio-temporal models. However, they are permanently isolated in traditional

Table 1

Descriptive statistics of our dataset.

	Occupancy	Temperature	Neighbors
Max.	100%	36 °C	12
Min.	0%	24 °C	0
Avg.	26.38%	28.50 °C	4.90
Std.	0.17	2.80	3.52

Max., Min., Avg., Std. represent the maximum value, minimum value, average value and standard deviation, respectively.

graph learning models, and information from other nodes cannot be propagated to these outliers. There is a total of 54,623 parking spaces, with an average of 941.78 spaces per parking lot, and each parking lot has electronic entry and exit recording facilities and property management services.

4.1.2. Weather data

Weather forecast data from the Guangzhou government¹ is introduced for classifying three scenarios (i.e. no rain, light rain, and heavy rain) according to the day's rainfall, as precipitation is one of the most important factors for traveling in southern China. In this month, there were 15 days without rain, 5 days with light rain, and 10 days with heavy rain. Three weather forecast scenarios are used for the division of the meta-training set and global attention computation to obtain better prediction results. As can be seen from Fig. 5, different weather can lead to different global attention, thus studies for different weather conditions are necessary.

The temperature data of Guangzhou is inputted into the model as time-series auxiliary information, as temperature has a specific impact on traffic travel. The sampling interval of temperature data is 30 min, which is processed into a 5-minute interval by linear interpolation. Since the locations of the weather stations and the parking lots do not coincide, we select the record of the nearest weather station for each parking lot as its auxiliary information. Note that real-time temperature data is used as an example and case study. The isomorphic inputs to our model data are not limited to temperature data. Information that affects parking behavior in real time, including traffic flow, economic development indicators, etc., can be used as inputs to the multivariate decoder.

Descriptive statistics of our dataset is shown in Table 1.

4.1.3. Data division

First, in all comparison models, the training and test sets are divided by 7:3, i.e., June 1 to June 21 for the training set and June 22 to June 30 for the test set. Second, in the Meta-HWFST model pre-trained by Reptile, there is no need to divide the support set and the query set. The meta-training, finetuning-training and test sets are divided by 5:2:3, i.e., June 1 to June 15 for the meta-training set, June 16 to June 21 for the finetuning training set and June 22 to June 30 for the test set. Third, in the Meta-HWFST model pre-trained by MAML, the data division is the same as the Meta-HWFST model pre-trained by Reptile, and the support set and the query set are divided by 7:3 in the meta-training set.

4.2. Comparison models

A statistical model and ten neural network models are selected as comparison models to analyze all the advantages of the proposed models.

(1) Least Absolute Shrinkage and Selection Operator (Lasso) [8]: A statistical regression model. It is a compression estimation method with

¹ <https://data.gz.gov.cn/>

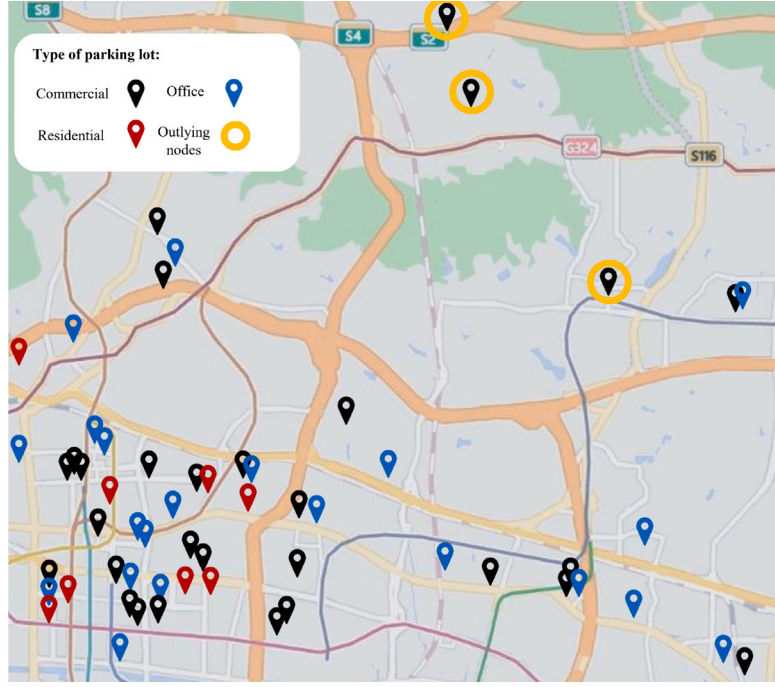
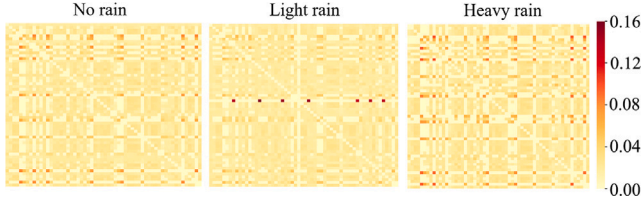


Fig. 4. Map of parking lots distribution in the dataset.

Fig. 5. Heatmaps of the global attention matrices A_j .

the idea of reducing the set of variables. It can compress the coefficients of the variables and make certain regression coefficients zero by constructing a penalty function, thus achieving variable selection.

(2) Fully connected neural networks (FCNN): A network that mimics the behavioral characteristics of animal neural networks. It processes information by adjusting the relationship between a large number of internal nodes interconnected with each other.

(3) Graph Attention Networks (GAT) [14]: An operation of aggregating features of neighboring vertices to the center vertex. The difference between GCN and GAT is that GCN utilizes the Laplacian matrix and GAT utilizes the attention coefficients. To some extent, GAT is stronger because the correlation between vertex features is better incorporated into the model.

(4) Long Short Term Memory (LSTM) [12]: A special kind of recurrent neural network. It can dynamically process the “memory” of the neural network to achieve excellent results in time series processing.

(5) Du-parking [10]: A deep learning parking occupancy prediction model for Baidu Map. This is a business model that has been widely used in major cities such as Shenzhen and Beijing. This model utilizes LSTM and consists of three major components modeling temporal closeness, period and current general influence.

(6) GT-LSTM [48]: A spatio-temporal ensemble network for traffic flow prediction. It captures the spatial dependence of traffic flow by using adaptive graph convolutional networks, as well as extracting traffic flow patterns using temporal convolutional networks and bi-directional LSTM networks.

(7) BEGAN [49]: A model combining GAT and LSTM, which is able to assign reasonable weights on the graph and learn spatio-temporal features.

(8) GCNN+LSTM [22]: A model combining GCN and LSTM, which not only learns about spatio-temporal relationship, but also utilizes convolution to incorporate temperature data.

(9) FourierGNN [50]: A model that proposes a novel pure graph perspective and uniformly considers spatio-temporal dynamics, allowing global information to be considered at once. The model achieves state-of-the-art in a number of spatio-temporal regression tasks.

(10) Global-LSTM: A model combining the global attention in the proposed model Meta-HWFST and LSTM, which can learn global correlations and time-series features.

(11) The proposed method pre-trained by MAML. To verify the performance of Reptile, the previously proposed meta-learning method MAML is also used for comparison.

Since GCNN+LSTM considers both spatio-temporal and temperature features, we set it as the baseline model.

4.3. Evaluation metrics

To make a full and fair comparison, four metrics, i.e., Root Mean Squared Error (RMSE), Relative Absolute Error (RAE), Mean Absolute Error (MAE), and R-Square Coefficient (R^2), are used. The calculation formulas are presented in Eq. (17).

$$\begin{aligned}
 \text{RMSE} &= \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \\
 \text{RAE} &= \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{\sum_{i=1}^N |y_i - \bar{y}|} \\
 \text{MAE} &= \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \\
 R^2 &= 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}
 \end{aligned} \tag{17}$$

where y_i and $\hat{y}_i$ are the observed and predicted value of sample i , respectively; $\bar{y}$ is the mean value of samples; and N is sample size.

4.4. Loss function and optimizer

In all the learning methods, a uniform loss function, Mean Square Error (MSE), is utilized, as shown in Eq. (18).

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (18)$$

In addition, a uniform optimizer, Adam, is used in all learning models.

4.5. Experimental setup

We conduct hyperparameter search experiments and select the most favorable hyperparameters for our proposed model. In Meta-HWFST, K (i.e., heads of the multi-head GAT) is set to 4, α in the global attention computation is set to 0.8, and the threshold δ to determine whether parking lots are adjacent to each other is set to 2 km, which is the maximum acceptable walking distance for most travelers. Note that the value of hyper-parameters needs to be set according to the actual scenario. In addition, the experiments are conducted to separately predict the parking occupancy in four intervals, namely 15, 30, 45, and 60 min (i.e., their time step values are 3, 6, 9, and 12, respectively, as the temporal resolution of the dataset is 5 min). Then, the max epoch of meta pre-training is set to 1000, and the max epoch of training for the comparison model and finetuning training is set to 600. Besides, the outer and the inner step size of meta-learning are set to 0.001 and 0.01. Finally, all the experiments are conducted on a Windows workstation with a NVIDIA RTX 4000 GPU, an Inter(R) Core(TM) i9-10900K CPU, and 64G RAM.

It is worth noting that for reproducibility, the code used by this paper are shared in Github, and downloadable from the link.²

4.6. Experiment result and discussion

The experiments are divided into three parts: (1) comparison experiments to demonstrate the proposed model ahead of other benchmark models, (2) ablation experiments to demonstrate the necessity of each module of the model, and (3) noise impulse experiments to illustrate the importance of each feature to validate the correctness of the learning.

4.6.1. Comparison experiment

As shown in Table 2, the evaluation metrics of compared models in four prediction intervals and the average value are summarized. The proposed methods outperform other methods in all four metrics with fewer forecasting errors. Moreover, compared to other methods, the proposed meta-learning spatio-temporal method, Meta-HWFST, on average, improves about 30.26% in RMSE, 33.53% in RAE, and 33.56% in MAE, and achieves the best fit in R^2 . Besides, Meta-HWFST, on average, improves by 37.64% over the currently applied commercial model, Du-parking. This excellent performance shows that modules can work together and make the model achieve state-of-the-art performance.

In particular, the proposed method Meta-HWFST improves by 29.81% over the baseline model, GCNN+LSTM, indicating that the proposed method is more efficient than the current convolutional method when fusing weather data. As a visual illustration, the 45-minute prediction results of a parking lot is plotted, as shown in Fig. 6, considering the average travel time in Guangzhou is close to 40 min [51]. It can be seen that the predicted curve of the proposed model is more consistent with the label curve than that of the baseline, i.e., GCNN+LSTM. More clearly, the error plot shows that the proposed model has more stable predictions during periods of occupancy fluctuations, such as during the evening rush hours.

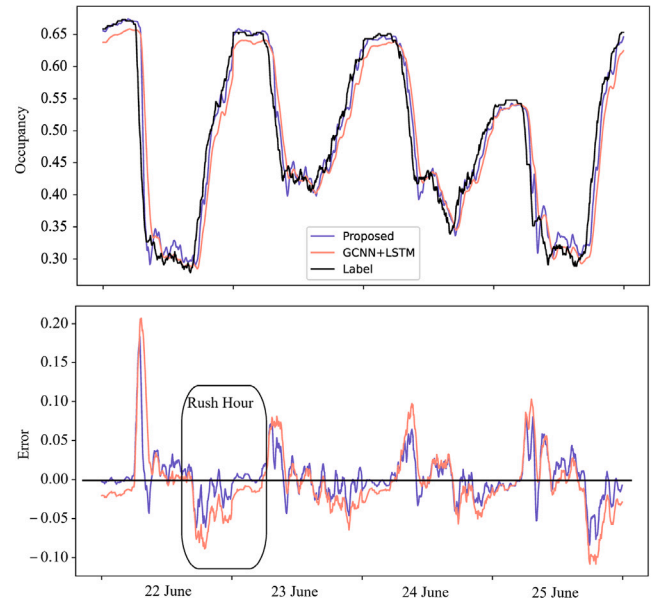


Fig. 6. Lineplots of the 45-min prediction results and errors for a parking lot.

As shown in Tables 2 and 3, the proposed method, Meta-HWFST performs better than other comparison models in four prediction tasks (i.e., 15 min, 30 min, 45 min, and 60 min), three types of weather (i.e., no rain, light rain, and heavy rain), and three types of parking lots (i.e., commercial, office, and residential). The proposed model performs well in different tasks, illustrating the wide applicability of this model.

In the comparison models, first, the poor performance of GAT and the superiority of BEGAN over GAT indicate that GAT cannot be used independently without considering the temporal effects in this prediction task. Second, BEGAN, Global-LSTM and GCNN+LSTM are better than LSTM, indicating the importance of learning the neighboring and global correlation and fusing parking-related environmental factors, respectively.

Compared to the proposed method pre-trained by MAML, Meta-HWFST, on average, improves about 6.90% in RMSE, 7.00% in RAE, and 6.98% in MAE, which demonstrates the superiority of Reptile over MAML in improving prediction accuracy. The training speed of different meta-learning methods are compared in Table 4. The training speed of Reptile is 21.12% faster than MAML, which is in line with the theory and shows the advantage of Reptile in training speed. While, normal training is very fast, the computational time consumed for normal training (i.e., fine-tuning) is 42% of that of reptile and 33% of that of MAML.

The model convergence speed is evaluated among some neural networks, i.e., LSTM, Du-parking, GT-LSTM, BEGAN, GCNN+LSTM, FourierGNN, *(MAML), and Meta-HWFST. Accordingly, a plot is drawn as shown in Fig. 7. Compared to other models, the loss of the proposed model achieves the minimum in the first 60 epochs. Moreover, a significant improvement is obtained by Meta-HWFST, whose loss curve starts smaller than other methods. Meanwhile, the proposed model pre-trained by Reptile has reached a better position than the proposed model pre-trained by MAML at the beginning of the loss curve. It shows that the proposed model equipped with weather-embedded Reptile can achieve a fast adaptation and convergence speed, and Reptile performs better than MAML in the pre-training process in our task.

4.6.2. Ablation experiment

To validate the effectiveness of the components in the proposed model (i.e., global attention, adjacent attention, temperature data fusion, multivariate decoder and weather-embedded meta-learning), an

² <https://github.com/kuanghx3/Meta-HWFST>

Table 2
The list of evaluation metrics of compared models.

	(15 min/30 min/45 min/60 min/Average)		
Metrics ($\times 10^{-2}$)	RMSE	RAE	
Lasso	3.66/4.29/4.99/5.69/4.66	20.08/23.00/26.13/29.30/24.63	
FCNN	4.40/5.24/6.01/6.72/5.59	18.30/22.08/25.70/29.14/23.80	
GAT	4.73/5.43/6.01/6.78/5.76	24.88/28.07/31.20/34.25/29.60	
LSTM	4.09/4.92/5.70/6.43/5.28	17.63/21.19/24.61/27.90/22.83	
Du-parking	4.09/4.86/5.64/6.43/5.22	17.27/20.68/24.04/27.35/22.33	
GT-LSTM	3.68/4.50/5.27/6.10/4.89	16.80/20.51/23.69/27.63/22.16	
BEGAN	3.68/4.48/4.32/5.12/4.88	16.58/19.86/23.56/27.22/21.81	
GCNN+LSTM	3.33/4.20/5.01/5.79/4.58	14.55/18.24/21.81/25.29/19.97	
FourierGNN	2.96/3.91/4.77/5.56/4.03	12.30/16.40/20.25/23.96/18.23	
Global-LSTM	2.62/3.56/4.32/5.12/3.90	12.33/16.06/19.66/23.09/17.79	
*(MAML)	2.16/3.08/3.96/4.72/3.48	9.00/13.24/17.06/20.74/15.01	
Proposed	1.91/2.76/3.78/4.52/3.24	7.91/11.82/16.38/19.75/13.96	
Metrics ($\times 10^{-2}$)	MAE	R ²	
Lasso	2.90/3.32/3.77/4.23/3.56	37.50/28.38/16.76/3.06/21.43	
FCNN	2.63/3.17/3.68/4.16/3.41	80.30/72.24/62.90/52.66/67.00	
GAT	3.57/4.02/4.47/4.90/4.24	28.40/17.50/5.34/−8.00/10.81	
LSTM	2.53/3.04/3.52/3.99/3.27	79.06/71.01/61.92/51.93/65.98	
Du-parking	2.48/2.96/3.44/3.99/3.20	79.50/72.29/63.98/54.59/67.59	
GT-LSTM	2.41/2.94/3.39/3.95/3.17	80.58/72.99/64.49/53.13/67.80	
BEGAN	2.38/2.85/3.37/3.89/3.12	78.69/72.03/62.46/50.61/65.95	
GCNN+LSTM	2.09/2.62/3.12/3.61/2.86	85.66/78.56/70.01/61.69/73.97	
FourierGNN	1.77/2.35/2.90/3.42/2.61	90.81/84.69/77.69/70.01/80.80	
Global-LSTM	1.77/2.30/2.81/3.30/2.55	85.70/80.93/68.29/59.51/73.61	
*(MAML)	1.29/1.90/2.44/2.96/2.15	94.27/88.60/80.93/71.55/83.84	
Proposed	1.14/1.69/2.34/2.82/2.00	95.16/90.43/81.35/75.59/85.63	

The bolded items are the best ones. Same below.

*(MAML) stands for the proposed method pre-trained by MAML. Same below.

Table 3
The list of evaluation metrics of compared models in groups.

	(No rain/Light rain/Heavy rain/Commercial/Office/Residual)		
Metrics ($\times 10^{-2}$)	RMSE	RAE	
Lasso	4.55/4.79/4.72/4.90/4.59/3.86	24.69/24.68/24.37/24.63/24.66/25.72	
FCNN	5.42/5.66/6.02/5.70/5.64/5.06	22.88/22.69/28.55/23.22/23.59/28.22	
GAT	5.64/5.88/5.99/6.07/5.54/5.16	28.83/29.01/33.10/30.24/27.65/34.23	
LSTM	5.13/5.44/5.53/5.47/5.27/4.57	22.11/22.05/26.43/22.45/22.80/25.73	
Du-parking	5.05/5.35/5.57/5.36/5.24/4.64	21.48/21.55/26.36/21.84/22.28/25.75	
GT-LSTM	4.74/5.05/5.12/5.12/4.79/4.19	21.48/21.46/25.47/21.88/21.85/25.36	
BEGAN	4.73/4.97/5.22/4.97/4.92/4.41	21.00/21.15/25.48/21.32/21.67/25.39	
GCNN+LSTM	4.44/4.67/4.88/4.73/4.58/4.01	19.24/19.30/23.45/19.77/19.86/22.13	
FourierGNN	4.18/4.35/4.60/4.43/4.30/3.78	17.55/17.44/21.67/17.94/18.02/21.02	
Global-LSTM	3.88/3.92/3.96/4.07/3.83/3.43	17.66/16.95/19.11/17.89/17.09/20.35	
*(MAML)	3.46/3.49/3.52/3.65/3.46/2.85	14.86/14.75/15.81/15.22/14.76/15.48	
Proposed	3.23/3.23/3.28/3.44/3.17/2.66	13.91/13.56/14.62/14.28/13.59/14.30	
Metrics ($\times 10^{-2}$)	MAE	R ²	
Lasso	3.46/3.70/3.62/3.68/3.57/3.06	−0.17/21.66/−2.85/53.86/−46.23/81.43	
FCNN	3.32/3.32/3.79/3.45/3.40/3.29	68.38/65.54/42.59/69.48/59.74/77.08	
GAT	4.18/4.25/4.40/4.49/3.98/3.99	12.44/6.68/−56.25/22.68/−26.13/64.75	
LSTM	3.21/3.23/3.51/3.33/3.29/3.00	67.68/64.58/39.53/73.93/49.51/80.41	
Du-parking	3.12/3.16/3.50/3.24/3.21/3.00	69.18/66.38/42.19/74.10/53.77/80.29	
GT-LSTM	3.12/3.14/3.39/3.25/3.15/2.96	69.56/66.67/39.51/75.47/52.69/79.62	
BEGAN	3.05/3.10/3.39/3.17/3.12/2.96	67.21/65.15/39.09/74.01/49.07/81.05	
GCNN+LSTM	2.79/2.83/3.12/2.94/2.86/2.58	75.28/72.97/51.09/77.74/64.87/84.26	
FourierGNN	2.55/2.55/2.88/2.66/2.60/2.45	81.46/79.59/66.46/81.20/78.19/86.21	
Global-LSTM	2.56/2.44/2.63/2.66/2.46/2.37	74.21/75.05/53.27/78.72/62.64/83.87	
*(MAML)	2.15/2.12/2.18/2.26/2.13/1.80	83.49/83.52/74.22/84.80/79.93/90.59	
Proposed	2.01/1.95/2.01/2.12/1.96/1.67	85.20/85.61/76.76/86.24/82.44/91.80	

All terms are the average of 4 prediction times.

Table 4
The training speed of different meta-training method.

	Finetuning	MAML	Reptile
Time (s/epoch)	0.280	0.838	0.661

ablation experiment with all 58 parking lots in all four prediction intervals is conducted, and the average result is shown in Table 5. Compared to the ablation models, Meta-HWFST, on average, improves

about 11.47% in RMSE, 12.83% in RAE, and 12.83% in MAE and achieves best fit in R², indicating the necessity of each part of the model. Moreover, it can be concluded that the global attention plays the most significant role with a 20.76% improvement and weather-embedded meta-learning with a 15.47% improvement ranks second. Besides, the 5.15% improvement brought by our proposed multivariate decoder indicates its important role. Furthermore, we discuss the computational complexity introduced by each module. Table 5 demonstrates the training speed of each model for the ablation experiment. It can be seen that each module brings a certain complexity to the model.

Table 5

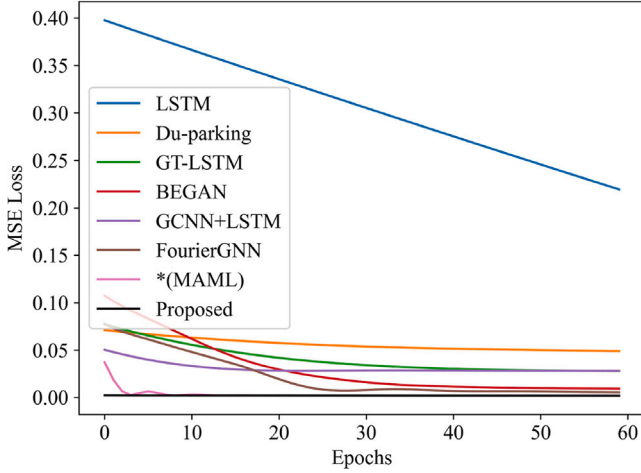
The list of evaluation metrics and training speed of ablation models.

Metrics ($\times 10^{-2}$)	RMSE	RAE	MAE	R ²	Training speed (s/epoch)
# Global attention	4.02	17.78	2.55	79.66	0.654
# Pre-training	3.77	16.66	2.39	73.12	0.280
# Temperature	3.61	15.72	2.25	83.85	0.662
# Adjacent attention	3.56	15.52	2.22	82.75	0.625
# Decoder	3.41	14.74	2.11	83.82	0.450
Full	3.24	13.96	2.00	85.63	0.661

Note that “#” denotes “without”.

All terms are the average of 4 prediction times.

Except for “# Pre-training”, training speed of the models denotes the pre-training speed.

**Fig. 7.** Plot of the first 60 training MSE loss in 45-min prediction task.

First, pre-training has the highest computational consumption because of its complex gradient descent process. Second, the multivariate decoder holds a high computational complexity because of the stepwise recurrent process of LSTM. Third, adjacent attention is required to determine attention and propagate information on the graph, but brings about only a weak decrease in training speed. It is worthwhile to recognize that some modules which introduce heterogeneous and real-time data (i.e., global attention and temperature) do not introduce significant computational complexity, because these processes barely change the existing network.

4.6.3. Hyperparameter sensitivity analysis

To evaluate the robustness of our model to hyperparameters, we compared the performance under different numbers of attention heads in GAT (i.e. K) and global attention fusion hyperparameter (i.e., α). As shown in Fig. 8, the performance of the model does not change drastically when the hyperparameters are changed, illustrating the robustness of the proposed model. Furthermore, our parameter sensitivity experiments illustrate the possible influence of the model by different settings. First, when the number of attention heads K is 4, the model has a good performance. When K is too small, there are not enough angles available for the model to focus on and performance is poor. When K increases, the amount of information provided to the model is too large and may bring about overfitting problems. We cannot deny that the performance of the model may be good when K continues to increase, because at this stage the attention mechanism can focus on a very large number of perspectives, but this brings an increase in computational complexity. Second, α is used to adjust the weighting of global auxiliary information and own information. We do not recommend setting α to a low value (e.g., below 0.5) because we believe that the node's own characteristics are very important in the prediction process. The experiment proves exactly that. Besides, a too high α may lead to underweighting of global information and its

role is diminished, thus resulting in a decrease in prediction accuracy. Our hyperparameter selection is based on this experiment, shown in Section 4.5.

4.6.4. Feature importance

Inspired by Partial Dependence Plots (PDP) [52], a noise impulse experiment is carried out to find out the importance of each occupancy feature and temperature feature. Standard normally distributed random noise is added to each occupancy feature and temperature feature of the test set. The more the MSE increases, the more important the feature is. The results are shown in Fig. 9. It can be learned that, first, the occupancy and temperature data closest to the current moment (i.e., t and $t-1$) are most heavily weighted in the forecast. Second, the occupancy features are over 10^2 times more important than the temperature features. Third, the temperature feature plays a more important role in short-term predictions (i.e., 15 min) than in long-term predictions. The above conclusions are in line with common sense and demonstrate the reasonableness of the attention allocation in the temporal decoder module and the correctness of the learning.

5. Conclusion and future work

Parking occupancy prediction allows drivers to know the usage of parking facilities at their destination in advance, and informs the formulation of smart parking guidance systems and pricing strategies. Moreover, spatio-temporal parking occupancy prediction has become a current research hotspot for its ability to provide information for regional management. However, there are still gaps in previous studies that make the predictions unsatisfactory. To solve the above problems simultaneously, a deep meta-learning approach, named Meta-HWFST, is proposed, which incorporates a global-adjacent attention module, a multivariate decoder module and a model pre-training module. First, to address the challenge of global data correlation, we propose an adaptive method for perceiving global attention by utilizing Pearson's correlation coefficient. We combine it with adjacent graph attention and propose a global-local dual-attention module to simultaneously perceive the patterns of urban parking lots. Second, improving the temporal learning ability and effectively embedding real-time data are ongoing research hotspots. We propose a multivariate decoder module that can input isomorphic temporal data to capture dynamics and reduce hysteresis, and improve long and short-time temporal learning capabilities through different lengths of LSTMs and temporal attention. Third, previous studies mainly used a convolution approach to consider parking-related factors, with the resulting problem of neglecting long-term effects. Accordingly, we developed Reptile to learn features under weather classification, which takes into account the long-time effects of heterogeneous weather. In particular, reptile provides a method for learning the inherent characteristics of data without loss of commonality.

As shown by the experiment results, the proposed approach outperforms other methods in three aspects. (1) The prediction errors are significantly reduced. The proposed method, Meta-HWFST, on average, improves 32.45% over compared models and 29.81% over our baseline model GCNN+LSTM, and achieves the best fit in R^2 . (2) Compared

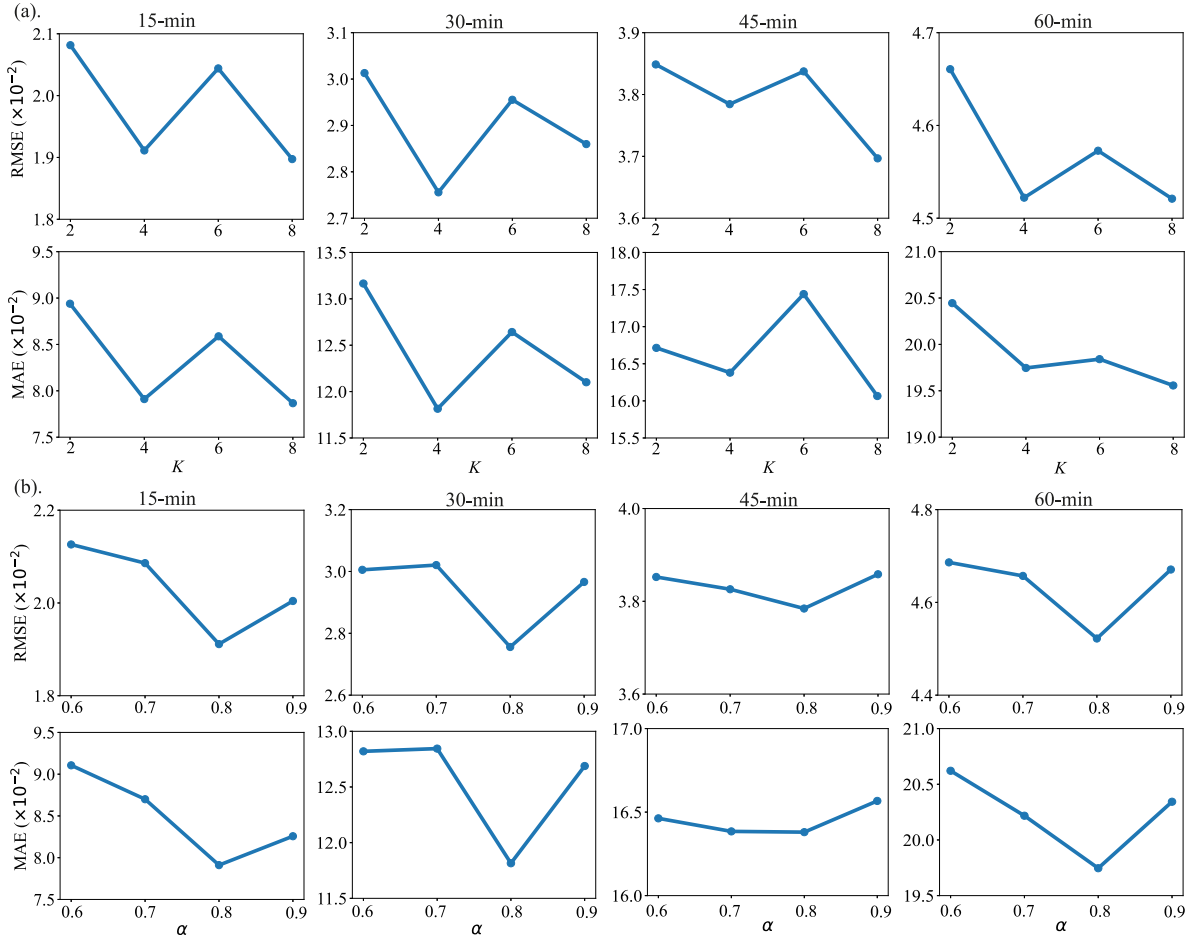


Fig. 8. Model performance with different hyperparameter values.

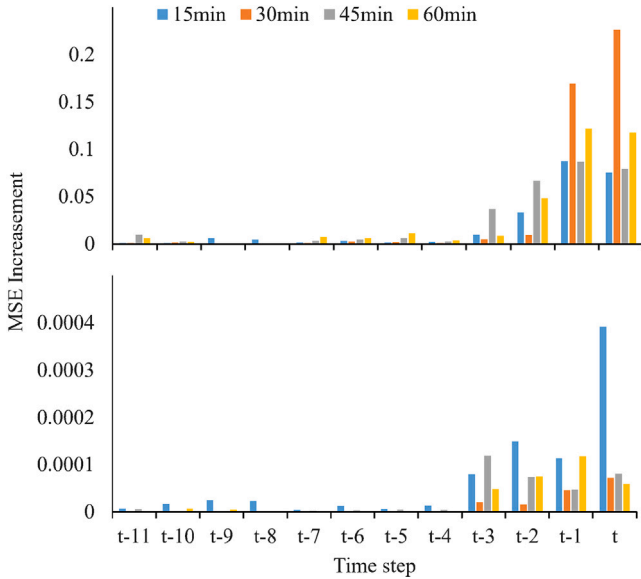


Fig. 9. Test MSE increase after adding noise. Upper: occupancy. Lower: temperature.

to other models, the training loss of Meta-HWFST starts at a lowest position and remains at a minimum. (3) The ablation experiment shows that Meta-HWFST, on average, improves about 12.38% and achieves the best fit in R^2 , indicating the necessity of each part of the model. Moreover, we also demonstrated through a noise impulse experiment that the model can correctly assign weights among features. Furthermore, our research brings some enlightening findings. (1) Parking behavior is affected by long periods of weather, and the data patterns in parking lots may be different under weather situations. A differentiated focus on weather information is beneficial for accurate prediction. (2) The ablation experiments demonstrate the importance of focusing on common node features globally. (3) Although dynamic auxiliary information is much less important than occupancy information, we cannot ignore the enhancement it brings to the prediction.

Apart from accurate prediction schemes, our research holds some managerial perspectives for governments and administrators. First, joint regional prediction and management is important for parking lots. Different managers operate different parking lots, but the government can guide demand through collaborative management, which can reduce cruising time and fuel waste. Second, in joint parking management, it is equally important to focus on the demand patterns of the same type parking lots and the common demand effects of neighboring parking lots. Third, the collaborative management of different information sources (e.g., weather, traffic flow, emergencies, etc.) at the governmental level allows for more effective prediction of urban transportation demand.

Future work can be conducted in three directions: (1) Other heterogeneous and real-time parking related factors can be incorporated into the learning model. These factors include spatial ones (e.g., road

network structures, public transportation stations, and surrounding facilities) and temporal ones (e.g., cycles, special public events, and peaks). Our model can be tested in other spatial contexts and dynamic factors. (2) Our proposed parking occupancy model can be integrated into intelligent transportation system. Through roadside information terminals, smart phones and other information dissemination platforms, accurate regional parking occupancy predictions may provide early warning for government management and optimal routes for travelers. (3) Downstream applications including smart navigation, dynamic pricing can be developed, in which the methods we propose can be applied to practical parking management and policy decisions.

CRedit authorship contribution statement

Haoxuan Kuang: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Kunxiang Deng:** Writing – original draft, Methodology. **Qiuxuan Wang:** Writing – original draft. **Wei Ye:** Methodology, Data curation. **Haohao Qu:** Writing – review & editing, Methodology. **Jun Li:** Writing – review & editing, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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